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Department of Computer Science & Engineering

Session: AUG-DEC 2025

UE24MA242A: Mathematics for Computer Science Engineers

Classwork Problems

#	UNIT_1 Questions	Exercise Number																				
1.	The article “Wind-Uplift Capacity of Residential Wood Roof-Sheathing Panels Retrofitted with Insulating Foam Adhesive” (P. Datin, D. Prevatt, and W. Pang, <i>Journal of Architectural Engineering</i>, 2011:144–154) presents tests in which air pressure was applied to roof-sheathing panels until failure. The pressure at failure for each panel was recorded, along with the type of sheathing, sheathing thickness, and wood species. The following table presents the results of four tests. <table><tr><th>Sheathing Type</th><th>Failure Pressure (kPa)</th><th>Thickness (mm)</th><th>Wood Species</th></tr><tr><td>5-ply plywood</td><td>2.63</td><td>11.9</td><td>Douglas Fir</td></tr><tr><td>Oriented strand board</td><td>3.69</td><td>15.1</td><td>Spruce-Pine-Fir</td></tr><tr><td>Cox plywood</td><td>5.26</td><td>12.7</td><td>Southern Yellow Pine</td></tr><tr><td>4-ply plywood</td><td>5.03</td><td>15.9</td><td>Douglas Fir</td></tr></table> Which data are numerical and which are categorical? Ans: - Numerical - Failure Pressure(kPa), Thickness(mm) These variables are measured with numbers and represent quantities - Categorical - Sheathing Type, Wood species These variables represent categories/groups and are not measured numerically	Sheathing Type	Failure Pressure (kPa)	Thickness (mm)	Wood Species	5-ply plywood	2.63	11.9	Douglas Fir	Oriented strand board	3.69	15.1	Spruce-Pine-Fir	Cox plywood	5.26	12.7	Southern Yellow Pine	4-ply plywood	5.03	15.9	Douglas Fir	Example 1.8
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2.	Each of the following processes involves sampling from a population. Define the population, and state whether it is tangible or conceptual. a. A chemical process is run 15 times, and the yield is measured each time. b. A pollster samples 1000 registered voters in a certain state and asks them which candidate they support for governor. c. In a clinical trial to test a new drug that is designed to lower cholesterol, 100 people with high cholesterol levels are recruited to try the new drug. d. Eight concrete specimens are constructed from a new formulation, and the compressive strength of each is measured. e. Quality engineers need to estimate the percentage of bolts manufactured on a certain day that meet a strength specification. At 3:00 in the afternoon they sample the last 100 bolts to be manufactured. Ans:	Exercise 1.1																				

	a) - Population: The set of all possible yields that could be obtained by running this chemical process under the same conditions. - Type: Conceptual, because the interest is in the distribution of all possible yields, which is theoretical b) - Population: All registered voters in that state. - Type: Tangible, the voters actually exist as a finite, identifiable group. c) - Population: All people with high cholesterol levels that the drug might be intended for. - Type: Tangible, such individuals exist in reality (though the trial may represent just a subset). d) - Population: All possible compressive strengths that could be observed if specimens were made with this formulation. - Type: Conceptual, we are considering the distribution of possible outcomes, not a fixed finite set. e) - Population: All bolts manufactured on that particular day. - Type: Tangible, the bolts actually exist and form a finite set.	
3.	A medical researcher wants to determine whether exercising can lower blood pressure. She recruits 100 people with high blood pressure to participate in the study. She assigns a random sample of 50 of them to pursue an exercise program that includes daily swimming and jogging. She assigns the other 50 to refrain from vigorous activity. She measures the blood pressure of each of the 100 individuals both before and after the study. a. Is this a controlled experiment or an observational study? Ans: In an observational study, the researcher only observes the subjects without assigning treatments. For example, comparing people who already exercise with those who do not. • In a controlled experiment, the researcher assigns subjects to different treatment groups under controlled conditions. Here: • The researcher randomly assigns 50 people to the exercise program and 50 to no vigorous activity. • This is a clear case of intervention and control. Hence, it is a controlled experiment!	Exercise 1.1
4.	In the article “Evaluation of Low-Temperature Properties of HMA Mixtures” (P. Sebaaly, A. Lake, and J. Epps, Journal of Transportation Engineering, 2002: 578–583), the following values of fracture stress (in megapascals) were measured for a sample of 24 mixtures of hot-mixed asphalt (HMA). 30 75 79 80 80 105 126 138 149 179 179 191 223 232 232 236 240 242 245 247 254 274 384 470 - Compute the mean, median, and the 5%, 10%, and 20% trimmed means - Find the mode and range - Find the first and third quartiles of the asphalt data - Find the 65th percentile of the asphalt data	Example 1.12

	Ans: a) $Mean = \sum Values / n = 5257 / 24 \approx \cancel{219.04 MPa} \quad 193. \sim$ $Median\ position = (n + 1) / 2 = (24 + 1) / 2 = 12.5$ $Median = Average\ of\ the\ 12th\ and\ 13th\ values$ $= (191 + 223) / 2 = 207\ MPa$ • $5\%\ Trimmed\ Mean = Mean\ of\ values\ from\ 2nd\ to\ 23rd\ observations:$ $Sum = 5202\ MPa, Mean = 5202 / 22 \approx 236.45\ MPa$ • $10\%\ Trimmed\ Mean = Mean\ of\ values\ from\ 3rd\ to\ 22nd\ observations:$ $Sum = 4644\ MPa, Mean = 4644 / 20 = 232.20\ MPa$ • $20\%\ Trimmed\ Mean = Mean\ of\ values\ from\ 6th\ to\ 19th\ observations:$ $Sum = 3207\ MPa, Mean = 3207 / 14 \approx 229.07\ MPa$ b) $Modes\ 80\ and\ 179\ as\ they\ appear\ twice$ $Range = Max - Min = 470 - 30 = 440\ MPa$ c) $First\ Quartile\ (Q1) - 25th\ percentile\ position = 0.25 \times (24+1) = 6.25$ $Q1 \approx value\ between\ 6th\ and\ 7th\ data\ points\ (average):$ $Q1 = \frac{105+126}{2} = \frac{231}{2} = 115.5\ MPa$ $Third\ Quartile\ (Q3) - 75th\ percentile\ position = 0.75 \times (24+1) = 18.75$ $Q3 \approx value\ between\ 18th\ and\ 19th\ data\ points\ (interpolated):$ $Q3 = \frac{242+245}{2} = \frac{487}{2} = 243.5\ MPa$ d) $65th\ Percentile$ $Position = 0.65 \times (24+1) = 16.25$ $This\ lies\ between\ 16th\ (236)\ and\ 17th\ (240)\ values:$ $65th\ percentile = \frac{236+240}{2}$ $= \frac{238}{2} = 119\ MPa$	
5.	In a sample of 20 men, the mean height was 178 cm. In a sample of 30 women, the mean height was 164 cm. What was the mean height for both groups put together? Ans: $Combined\ mean = \frac{(n1 \times mean1) + (n2 \times mean2)}{n1 + n2}$ $= \frac{(20 \times 178) + (30 \times 164)}{20 + 30} = \frac{3560 + 4920}{50} = \frac{8480}{50} = 169.6\ cm$	Exercise 1.2
6.	Why can't the average of nominal data be computed? Ans:	Data types

	The average of nominal data cannot be computed because nominal data represents categories or labels without any inherent numerical value or order. Since nominal data is qualitative and categorical in nature, the values are just names or labels (e.g., gender, color, nationality) and do not have a meaningful numeric relationship that would allow arithmetic operations like addition or averaging.	
7.	Is it possible to sort ordinal data meaningfully but not perform arithmetic operations on it? Why or why not? Ans: Yes. - Reason: Ordinal data represent categories with a meaningful order or ranking (e.g., “poor,” “fair,” “good,” “excellent”), so you can sort and rank them. - However, the intervals between categories are not necessarily equal or known, which means you cannot perform meaningful arithmetic operations like addition, subtraction, or averaging on ordinal values. - You can use median or rank-based statistics, but operations assuming equal intervals (like mean) are not valid.	Data types
8.	Why is temperature in Celsius considered interval data and not ratio data? Ans: Temperature in Celsius is interval data because: - It has equal intervals between values (e.g., the difference between 10°C and 20°C is the same as between 20°C and 30°C). - But it has no true zero point—0°C does not mean “no temperature” but is just the freezing point of water. - Because of this lack of a true zero, you cannot say that 20°C is twice as hot as 10°C, so it is not ratio data. - Ratio data require a meaningful absolute zero, like weight or height.	Data types
9.	You are asked to analyze customer satisfaction scores on a 5-point scale (1 to 5). Can you treat these scores as interval data? Justify. Ans: Generally, no or cautiously. - Customer satisfaction scores are ordinal data because the points reflect a rank order (e.g., 5 is better than 4), but the difference between points may not be equal or perceived equally by participants. - Without evidence that intervals between points are equal, treating them as interval data (for calculating means or standard deviations) can be misleading. - However, in some practical situations, researchers treat Likert-type scales as interval data to perform analyses like averaging, but this is an assumption and not strictly correct.	Data types
10.	A student records eye color and test scores. Which is qualitative and which is quantitative data? How does this affect analysis? Ans: - Eye color: Qualitative (categorical) data because it describes categories (e.g., blue, brown, green) without numeric measurement. - Test scores: Quantitative data because they are numerical measurements that can be meaningfully added, averaged, and analyzed with math operations.	Data types

	- Effect on analysis: - Qualitative data is analyzed using frequencies, mode, or chi-square tests. - Quantitative data can be analyzed with means, standard deviations, correlations, regressions, etc.	
11.	A piston is placed inside a cylinder. The clearance is the distance between the edge of the piston and the wall of the cylinder and is equal to one-half the difference between the cylinder diameter and the piston diameter. Assume the piston diameter has a mean of 80.85 cm with a standard deviation of 0.02 cm. Assume the cylinder diameter has a mean of 80.95 cm with a standard deviation of 0.03 cm. Find the mean clearance. Assuming that the piston and cylinder are chosen independently, find the standard deviation of the clearance. Ans: Piston diameter D_p: mean $\mu_p = 80.85$ cm, std dev $\sigma_p = 0.02$ cm Cylinder diameter D_c: mean $\mu_c = 80.95$ cm, std dev $\sigma = 0.03$ cm Clearance $C = \frac{D_c - D_p}{2}$ Mean Clearance using the linearity of expectation $\mu C = \frac{\mu_c - \mu_p}{2} = \frac{80.95 - 80.85}{2} = \frac{0.10}{2} = 0.05$ cm Standard Deviation of Clearance σC: $Var(D_c - D_p) = Var(D_c) + Var(D_p) = \sigma_c^2 + \sigma_p^2$ $Var(C) = (\frac{1}{2})^2 * (\sigma_c^2 + \sigma_p^2) = \frac{1}{4} (\sigma_c^2 + \sigma_p^2)$ $\sigma C = \frac{1}{2} \sqrt{\sigma_c^2 + \sigma_p^2} = \frac{1}{2} \sqrt{0.03^2 + 0.02^2} = \frac{1}{2} \sqrt{0.0009 + 0.0004} = \frac{1}{2} \sqrt{0.0013} \approx 0.03606$ $\sigma C = \frac{0.03606}{2} = 0.01803$ cm	Example 2.50
12.	If X and Y are independent random variables with means $\mu_X = 9.5$ and $\mu_Y = 6.8$, and standard deviations $\sigma_X = 0.4$ and $\sigma_Y = 0.1$, find the means and standard deviations of the following: $3X$ $Y - X$ $X + 4Y$ Ans: μ of nX is $n\mu_X$, σ of nX is $n\sigma_X$ - Mean: $\mu_{3X} = 3\mu_X = 3 \times 9.5 = 28.5$ Std. deviation: $\sigma_{3X} = 3 \times \sigma_X = 3 \times 0.4 = 1.2$ - Mean: $\mu_Y - X = \mu_Y - \mu_X = 6.8 - 9.5 = -2.7$ Std. deviation (Since X and Y are independent): $\sigma_{X-Y} = \sqrt{\sigma_X^2 + \sigma_Y^2} = \sqrt{0.4^2 + 0.1^2} = \sqrt{0.16 + 0.01} = \sqrt{0.17} \approx 0.4123$	Exercise 2.5

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$$C = \frac{1}{2} \left[\overset{\text{cylinder diam}}{x} - \underset{\text{piston diam}}{y} \right]$$

$$E[C] = 0.5 E[x] - 0.5 E[y]$$

$$\text{var}[C] = 0.5^2 \text{var}[x] + 0.5^2 \text{var}[y]$$

$$\text{var}(x) = (\text{std deviation})^2$$

$$SD[C] = \sqrt{\text{var}(C)}$$

	d) <i>Mean:</i> $\mu X + 4Y = \mu X + 4\mu Y = 9.5 + 4 \times 6.8 = 9.5 + 27.2 = 36.7$ <i>Std. deviation:</i> $\sigma X + 4Y = \sqrt{(\sigma X)^2 + (4\sigma Y)^2} =$ $\sqrt{(0.4)^2 + (4 \times 0.1)^2} = \sqrt{0.16 + 0.16} = \sqrt{0.32} \approx 0.566$	
13.	A laminated item is composed of five layers. The layers are a simple random sample from a population whose thickness has mean 1.2 mm and standard deviation 0.04 mm. a. Find the mean thickness of an item. b. Find the standard deviation of the thickness of an item. Ans: Given: Each layer's thickness (X) is a random sample from a population Mean (μ_X) = 1.2mm Standard deviation (σ_X) = 0.04 mm Total thickness (S) = $X_1 + X_2 + X_3 + X_4 + X_5$ Each X_i is independent and identically distributed Mean thickness of the item (mean of sum = sum of means) $\mu_S = 5\mu_X = 5 \times 1.2 = 6.0 \text{ mm}$ Std. deviation of the thickness of the item (for independent layers, variance of sum = sum of variances) $Var(S) = \sum_{i=1}^5 Var(X_i) = 5\sigma_X^2 = 5 \times (0.04)^2 = 5 \times 0.0016 = 0.008$ $\sigma_S = \sqrt{Var(S)} = \sqrt{0.008} \approx 0.0894 \text{ mm}$	Exercise 2.5
14.	Suppose that we extract an individual at random from a population whose members have an average income of \$40,000, with a standard deviation of \$20,000. What is the probability of extracting an individual whose income is either less than \$10,000, or greater than \$70,000? Ans: Mean income $\mu = \\$40,000$ Std. deviation $\sigma = \\$20,000$ We want the probability $P(X < \\$10,000 \text{ or } X > \\$70,000)$ Distance from the mean: $\\$10,000 - \\$40,000 = \\$70,000 - \\$40,000 = \\$30,000$ in terms of std deviation $\frac{\\$30,000}{\\$20,000} = 1.5\sigma$ So, we are looking for the probability that X is at least 1.5 standard deviations away from the mean: $P(X - \mu \geq 1.5\sigma)$ <i>Handwritten notes:</i> dispersion: 30,000 $K = \frac{30,000}{20,000} = 1.5$ $\Rightarrow 1.5\sigma$	Chebyshev's inequality

Chebyshev's inequality states:

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

$$P(|X - \mu| \geq 1.5\sigma) \leq \frac{1}{1.5^2} = \frac{1}{2.25} \approx 0.444$$

The probability that an individual's income is less than \$10,000 or greater than \$70,000 is at most 44.4%

15. Suppose the average number of requests a web server receives per minute is 100, with a standard deviation of 10. What is the minimum percentage of time the number of requests will be between 80 and 120

Chebyshev's inequality

Ans:

Mean = 100, std deviation = 10, Range: 80-120

Number of std deviations from the mean:

$$80 = 100 - 20 = 100 - 2 \times 10$$

$$120 = 100 + 20 = 100 + 2 \times 10$$

So, 80 and 120 are both 2 standard deviations away from the mean

Chebyshev's inequality states that at least $1 - \frac{1}{k^2}$ of the data lies within k

k standard deviations of the mean (for any distribution)

Here, $k=2$

$$1 - \frac{1}{2^2} = 1 - \frac{1}{4} = \frac{3}{4} = 0.75$$

At least 75% of the time, the number of requests will be between 80 and 120

16. A program has an expected execution time of 10 seconds with a standard deviation of 2 seconds. Using Chebyshev's inequality, what is the probability that the program will take longer than 16 seconds to execute?

Chebyshev's inequality

Ans:

Mean execution time, $\mu = 10$ seconds

Std deviation $\sigma = 2$ seconds

We want $P(X > 16)$

Step 1: Express in terms of deviations from the mean $16 - 10 = 6$ seconds

Calculate how many std. deviations above the means:

$$k = \frac{16 - 10}{2} = 3$$

Step 2: Apply Chebyshev's inequality

Since we want $P(X > 16)$ which is

is same as $P(|X - \mu| \geq k\sigma) = \frac{1}{k^2}$

$\Rightarrow P(|X - \mu| < k\sigma) = 1 - \frac{1}{k^2}$

for any other question

$$P(X - \mu \geq 3\sigma) \leq P(|X - \mu| \geq 3\sigma) \leq \frac{1}{3^2} = \frac{1}{9} \approx 0.1111$$

The probability that the program takes more than 16 seconds to execute is at most 11.11% (upper bound)

17.

As part of a quality-control study aimed at improving a production line, the weights (in ounces) of 20 bars of soap are measured. The results are as follows, sorted from smallest to largest.

11.6 12.6 12.7 12.8 13.1 13.3 13.6 13.7 13.8 14.1
14.3 14.3 14.6 14.8 15.1 15.2 15.6 15.6 15.7 15.8

Construct a normal probability plot for these data. Do these data appear to come from an approximately normal distribution?

Ans:

Calculate expected normal quantiles (theoretical quantiles):

- For each observation i in the sorted data ($i=1,2,\dots,20$), calculate the plotting position:

$$p_i = \frac{i - 0.5}{n} \text{ where } n = 20$$

Find the corresponding z-scores (quantiles of standard normal distribution) for these probabilities p_i

Plot the observed data values (y_i) against the theoretical z-scores (x_i)

If the data points fall approximately on a straight line, the data can be considered approximately normal.

Quick calculation example for first and last data points:

- For $i = 1$:

$$p_1 = \frac{1 - 0.5}{20} = \frac{0.5}{20} = 0.025$$

z-score for 0.025 $\approx$ -1.96

- For $i = 20$:

$$p_{20} = \frac{20 - 0.5}{20} = \frac{19.5}{20} = 0.975$$

z-score for 0.975 $\approx$ +1.96

If you plot these points and see an approximate straight line, the weights appear to be normally distributed.

Exercise
4.10

18.	Bottles filled by a certain machine are supposed to contain 12 oz of liquid. In fact the fill volume is random with mean 12.01 oz and standard deviation 0.2 oz. a. What is the probability that the mean volume of a random sample of 144 bottles is less than 12 oz? b. If the population mean fill volume is increased to 12.03 oz, what is the probability that the mean volume of a sample of size 144 will be less than 12 oz? Ans: a) Population mean $\mu = 12.01\text{oz}$ Population std deviation, $\sigma = 0.2\text{ oz}$ Sample size, $n = 144$ By CLT, the sampling distribution of the mean $\bar{X}$ is approximately normal with Mean: $\mu_{\bar{X}} = \mu$ Standard deviation (standard error): $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{0.2}{\sqrt{144}} = \frac{0.2}{12} = 0.01667$ Part (a): Probability mean volume of sample < 12 oz (with $\mu = 12.01$) Calculate the z-score: $z = \frac{12 - 12.01}{0.01667} = \frac{-0.01}{0.01667} \approx -0.6$ Using z-tables or normal distribution calculators, $P(\bar{X} < 12) = P(Z < -0.6) \approx 0.2743$ So, the probability is about 27.43%. Part (b): μ increases to 12.03 oz, find $P(\bar{X} < 12)$: $z = \frac{12 - 12.03}{0.01667} = \frac{-0.03}{0.01667} \approx -1.8$ Using z-tables: $P(\bar{X} < 12) = P(Z < -1.8) \approx 0.0359$ So, the probability is about 3.59%.	Exercise 4.11
19.	A 500-page book contains 250 sheets of paper. The thickness of the paper used to manufacture the book has mean 0.08 mm and standard deviation 0.01 mm. a. What is the probability that a randomly chosen book is more than 20.2 mm thick (not including the covers)? b. What is the 10th percentile of book thicknesses? c. Someone wants to know the probability that a randomly chosen page is more than 0.1 mm thick. Is enough information given to compute this probability? If so, compute the probability. If not, explain why not.	Exercise 4.11

Ans:

Number of sheets $n = 250$

Thickness of paper

$$\mu = 0.08\text{mm}, \sigma = 0.01\text{ mm}$$

a) Probability the book thickness is more than 20.2 mm

Total book thickness (excluding covers) is the sum of thicknesses of 250 sheets.

By the Central Limit Theorem:

Mean total thickness:

$$\mu_{total} = n \times \mu = 250 \times 0.08 = 20\text{ mm}$$

Standard deviation of total thickness:

$$\sigma_{total} = \sqrt{n} \times \sigma = \sqrt{250} \times 0.01 \approx 15.81 \times 0.01 = 0.1581\text{ mm}$$

Calculate the z-score for 20.2 mm:

$$z = \frac{20.2 - 20}{0.1581} \approx \frac{0.2}{0.1581} \approx 1.265$$

Using standard normal distribution tables,

$$P(X > 20.2) = 1 - P(Z \leq 1.265) \approx 1 - 0.8962 = 0.1038$$

The probability that the book is thicker than 20.2 mm is approximately 10.38%

b) Find the 10th percentile of book thicknesses

The 10th percentile corresponds to $z = -1.28$ (from standard normal tables)

Using the z-score formula:

$$X = \mu_{total} + z \times \sigma_{total} = 20 + (-1.28) \times 0.1581 \approx 20 - 0.202 = 19.798\text{ mm}$$

The 10th percentile is approximately 19.8mm

c) Probability a randomly chosen page is more than 0.1mm thick

We are asked for $P(\text{page thickness} > 0.1)$

Give the page thickness mean $\mu = 0.08\text{mm}$ and std deviation $\sigma = 0.01\text{mm}$

Assume the page thickness follows a normal distribution (usually reasonable for measurement data)

Calculate the z-score:

→ directly use population parameter here

→ find "0.1000" (0.1003) in Z table and find Z value

→ or just use INVER5-Normal in calc

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let the book have pages

$x_1, x_2, \dots, x_{250}$

from CLT sum

we have

$$X \sim N(n\mu, n\sigma^2)$$

where μ, σ^2 are population parameters

$$\Rightarrow \bar{x} = n\mu = 250 \times 0.08 = 20 \text{ mm}$$

$$S = \sqrt{n} \sigma = \sqrt{250} \times 0.01 = 0.1581$$

$\rightarrow$ now proceed with normal steps.

$$z = \frac{0.1 - 0.08}{0.01} = \frac{0.02}{0.01} = 2$$

Find $P(X > 0.1)$:

$$P(Z > 2) = 1 - P(Z \leq 2) = 1 - 0.9772 = 0.0228$$

Yes, enough information is given. The probability that a randomly chosen page is thicker than 0.1 mm is approximately 2.28%.

20.

A commuter encounters four traffic lights each day on the way to work. Let X represent the number of these that are red lights. The probability mass function of X is as follows.

x	0	1	2	3	4
$P(X = x)$	0.1	0.3	0.3	0.2	0.1

What is the probability that in a period of 100 days, the average number of red lights encountered is more than 2 per day?

Ans:

Step 1: Find the mean of

X (expected number of red lights per day):

$$\mu = \sum x \cdot P(X=x) = 0 \times 0.1 + 1 \times 0.3 + 2 \times 0.3 + 3 \times 0.2 + 4 \times 0.1$$

$$= 0 + 0.3 + 0.6 + 0.6 + 0.4 = 1.9$$

Step 2: Find the variance of X :

First calculate $E(X^2)$: $\sum$

$$x^2 \cdot P(X = x) = 0^2 \cdot 0.1 + 1^2 \cdot 0.3 + 2^2 \cdot 0.3 + 3^2 \cdot 0.2 + 4^2 \cdot 0.1$$

$$= 0 + 0.3 + 4 \times 0.3 + 9 \times 0.2 + 16 \times 0.1 = 0.3 + 1.2 + 1.8 + 1.6 = 4.9$$

Variance:

$$\sigma^2 = E(X^2) - \mu^2 = 4.9 - (1.9)^2 = 4.9 - 3.61 = 1.29$$

Standard deviation:

$$\sigma = \sqrt{1.29} \approx 1.135$$

Step 3: Use CLT for sample mean over 100 days

Sample size $n = 100$

Exercise
4.11

Mean of sample mean:

$$\mu_{\bar{X}} = \mu = 1.9$$

Standard deviation of sample mean (standard error):

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{1.135}{\sqrt{100}} = \frac{1.135}{10} = 0.1135$$

Step 4: Find the probability average number of red lights per day is more than 2

Calculate the z-score for $\bar{X} = 2$:

$$z = \frac{2 - 1.9}{0.1135} = \frac{0.1}{0.1135} \approx 0.88$$

Using standard normal tables or calculator:

$$P(\bar{X} > 2) = P(Z > 0.88) = 1 - P(Z \leq 0.88) \approx 1 - 0.8106 = 0.1894$$

The probability that the average number of red lights per day over 100 days is more than 2 is approximately 18.94%.

21.

The time taken by an automobile mechanic to complete an oil change is random, with mean 29.5 minutes and standard deviation 3 minutes.

- What is the probability that 50 oil changes take more than 1500 minutes?
- What is the probability that a mechanic can complete 80 or more oil changes in 40 hours?

g ~ N()

Ans:

Calculate using CLT

- Mean time per oil change, $\mu = 29.5$ minutes
- Standard deviation, $\sigma = 3$ minutes

- Probability that 50 oil changes take more than 1500 minutes

Total time for 50 oil changes: $T = \sum_{i=1}^{50} X_i$

By CLT, T is approximately normally distributed with

Mean: $\mu_T = 50 \times 29.5 = 1475$ minutes

Std deviation: $\sigma_T = \sqrt{50} * 3 = 3 * 7.071 = 21.213$ minutes

Find the z-score:

$$z = \frac{1500 - 1475}{21.213} = \frac{25}{21.213} \approx 1.18$$

- Using standard normal table,

$$P(T > 1500) = P(Z > 1.18) = 1 - P(Z \leq 1.18) \approx 1 - 0.8810 = 0.1190$$

Probability is approximately 11.9%

Exercise
4.11

	b) Probability that a mechanic completes 80 or more oil changes in 40 hours 40 hrs = 2400 minutes Let N = no. of oil changes completed in 2400 minutes each oil change time X_i has mean 29.5, sd 3 We approximate N by considering the total time S_n for n oil changes $P(N \geq 80) = P(S_{80} \leq 2400)$ By CLT: Mean for 80 oil changes: $\mu_{S_{80}} = 80 \times 29.5 = 2360 \text{ minutes}$ Standard deviation: $\sigma_{S_{80}} = \sqrt{80} \times 3 = 3 \times 8.944 = 26.833 \text{ minutes}$ Calculate z-score: $z = \frac{2400 - 2360}{26.833} = \frac{40}{26.833} \approx 1.49$ Find probability: $P(S_{80} \leq 2400) = P(Z \leq 1.49) \approx 0.9319$ The probability that the mechanic completes 80 or more oil changes in 40 hours is approximately 93.19%	
22.	There are 100 men on a plane. Let X_i be the weight (in pounds) of the ii-th man, with X_i's being i.i.d., $E[X_i] = \mu = 170$, and $\sigma X_i = 30$. Find the probability that the total weight exceeds 18,000 pounds. Ans: Number of men, $n=100$ Each man's weight X_i is i.i.d with mean $\mu = 170$ pounds and std deviation $\sigma = 30$ pounds Total weight $S = \sum_{i=1}^{100} X_i$ Find $P(S > 18,000)$	CLT

	Step 1: Mean and standard deviation of total weight S $\mu_S = n \times \mu = 100 \times 170 = 17,000$ $\sigma_S = \sqrt{n} \times \sigma = \sqrt{100} \times 30 = 10 \times 30 = 300$ Step 2: Calculate z-score for $S = 18,000$ $z = \frac{18,000 - 17,000}{300} = \frac{1,000}{300} = 3.33$ Step 3: Use standard normal distribution to find $P(S > 18,000)$ $P(S > 18,000) = P(Z > 3.33)$ Using z-tables or calculator: $P(Z > 3.33) \approx 1 - 0.99957 = 0.00043$ The probability that the total weight exceeds 18,000 pounds is approximately 0.043% (very small).	
23.	Generating Poisson Variates Random Numbers: 0.25, 0.70, 0.15, 0.90, 0.40 Goal: Generate 3 Poisson variates ($\lambda = 0.2$) Ans: $\alpha = 0.2, e^{-0.2} = 0.8187$ Variate 1 $n=0, P=1$ $R_1 = 0.25, P = 1 * 0.25$ $0.25 < e^{-0.2} \Rightarrow \text{Accept } N=0$ Variate 2 $n=0, P=1$ $R_2 = 0.70, P = 1 * 0.70$ $0.70 < e^{-0.2} \Rightarrow \text{Accept } N=0$ Variate 3 $n=0, P=1$ $R_3 = 0.15, P = 1 * 0.15$ $0.15 < e^{-0.2} \Rightarrow \text{Accept } N=0$ Hence the three variates are 0,0 and 0.	Random variate generation

24.	Generating Poisson Variates Random Numbers: 0.81, 0.73, 0.55, 0.21, 0.98 Goal: Generate 3 Poisson variates ($\lambda = 0.7$) Ans: $\alpha = 0.7, e^{-0.7} = 0.4965$ Variate 1 $n=0, P=1$ $R_1 = 0.81, P = 1 * 0.81$ $0.81 > e^{-0.7} \Rightarrow \text{Reject } n=0$ $n=1, P = 0.81*0.73$ $R_2 = 0.73, P = 0.5913$ $0.5913 > e^{-0.7} \Rightarrow \text{Reject } n=1$ $n=2, P = 0.5913*0.55$ $R_3 = 0.55, P = 0.325$ $0.325 < e^{-0.7} \Rightarrow \text{Accept } n=2$ Variate 2 $n=0, P=1$ $R_4 = 0.21, P = 1 * 0.21$ $0.55 < e^{-0.7} \Rightarrow \text{Accept } n=0$ Variate 3 $n=0, P=1$ $R_4 = 0.98, P = 1 * 0.98$ $0.21 > e^{-0.7} \Rightarrow \text{Reject } n=0$ Hence the 3 variates are 2, 0 and 0	Random variate generation
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